

Summer Internship Programme – PGDM – ISME25-27

Assignment Report | Steel & Construction Division

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**Organisation: Hindupur steels and
alloys (P) Ltd.**

Period: SIP Week 1&2

1. Introduction to Internship / Summer Training

This internship was undertaken at Hindupur steels and alloys (P) Ltd. , a TMT bar manufacturing plant, as part of an industrial training programme in the Steel & Construction sector. The training spanned from April 2 to April 13, 2026, covering two progressive weeks of hands-on exposure to the complete TMT bar production process — from raw material procurement and elemental testing, through smelting and billet casting, to rolling, quenching, and final quality control testing.

The Grade Testing Laboratory is an integral in-house quality control unit within the Hindupur steels and alloys (P) Ltd. plant. It conducts mandatory tensile testing of produced rebar every 30 minutes during active production runs of 8 mm, 12 mm, and 20 mm diameter TMT bars, ensuring each batch meets IS 1786 mechanical property requirements before dispatch. The internship thus provided end-to-end exposure — from the raw material stage at the Spectro Lab and smelter, all the way through to finished product verification at the Grade Testing Lab.

2. Relevance of Topic

TMT (Thermo-Mechanically Treated) bars are the single most critical material in reinforced concrete construction — used in buildings, bridges, dams, and roads across India. The relevance of this internship spans several key dimensions:

- **Structural Safety:** Understanding tensile strength, yield stress, and elongation % is essential for ensuring that rebars meet the mechanical property requirements of IS 1786 : 2008. The observed sample (20 mm bar) recorded a Yield Stress of 542 N/mm² and a Tensile Strength of 644 N/mm², confirming Fe 500 grade compliance.
- **Quality at Source:** The entire production chain at Hindupur steels and alloys (P) Ltd. relies on real-time Spectro Lab analysis (results in 5–10 seconds) to verify elemental composition of raw materials before any large-scale melting is commenced.
- **Seismic Resilience:** The Ring Test (Etching Test) reveals the three-zone microstructure — Tempered Martensite outer ring, Bainite transition zone, and Ferrite-Pearlite core — whose balance determines both the strength and ductility of the bar. This is directly relevant to earthquake-resistant construction governed by IS 1893 and IS 13920.
- **Industry Economics:** Power cost calculations observed during smelter operations (3.89 lakh per 18-tonne billet batch at 6/unit) illustrate the real cost pressures that drive quality-

efficiency decisions on the production floor.

- Standards Compliance: IS 1786 tolerance ranges ($\pm 7\%$ for 8–10 mm, $\pm 5\%$ for 12–16 mm, $\pm 3\%$ for 20 mm and above) directly impact procurement, quality rejection, and structural design calculations.

3. Theoretical Background

3.1 TMT Bar Manufacturing & Microstructure

TMT bars are produced by passing hot-rolled steel through a quenching water-jet system immediately after the rolling mill. The outer surface cools rapidly, forming a hard martensite shell, while the core remains hot and ductile. Upon air cooling, the residual heat from the core tempers the outer martensite (self-tempering), resulting in three distinct microstructural zones:

Zone	Crystal Structure	Appearance	Function
Tempered Martensite (outer ring)	Fine lath martensite	Dark / Black	High tensile strength
Transition Zone (middle band)	Bainite / Mixed	Medium / Brown	Toughness & shock absorption
Ferrite-Pearlite Core (inner)	Ferrite + Pearlite (soft)	Light / White	Ductility & flexibility

3.2 Ring Test (IS 1786 : 2008) — Etching Test

The Ring Test verifies the microstructure of a TMT bar by cutting a cross-section, polishing it smooth, and applying 5–10% Nital solution (nitric acid in ethanol) for 15–30 seconds. The chemical etches each zone differently: the martensite ring appears dark/black, the transition zone medium-brown, and the ferrite-pearlite core appears light. A continuous, unbroken dark outer ring constitutes a PASS; any break or unevenness in the ring indicates an uneven quench (FAIL), and the bar must be rejected.

3.3 Tensile Testing & Mechanical Properties

Tensile testing is conducted every 30 minutes during production. Key parameters measured include Yield Load, Tensile Load, Cross-Sectional Area (CSA), Yield Stress, Tensile Strength,

T.S./Y.S. ratio, and Elongation %. The sample recorded during training (20 mm bar equivalent): Yield Stress 542 N/mm², Tensile Strength 644 N/mm², T.S./Y.S. ratio 1.20, and Elongation 18–10% — all consistent with Fe 500D grade requirements under IS 1786. The weight formula $D^2/162$ (kg/m) is used to cross-check dimensional compliance, with IS 1786 tolerances of $\pm 7\%$ (8–10 mm), $\pm 5\%$ (12–16 mm), and $\pm 3\%$ (20 mm and above).

3.4 Raw Material Testing & Smelter Operations

Before any raw material (scrap iron, sponge iron, pig iron, gypsum) enters the furnace, a lollipop-shaped sample is prepared, polished flat, and tested in the Spectro Lab (precision ± 10 –15 units, results in 5–10 seconds). If results fail, the material is rejected or sent to the Chemical Lab for 3-hour confirmatory analysis. The smelter processes a minimum of 15–20 tonnes per cycle, producing 18-tonne billet batches (90% yield) at a melt temperature of 1380°C, consuming approximately 722 kWh per tonne. The molten steel is then cast into billets by the CCM (Continuous Casting Machine) unit, completing the raw-material-to-billet transformation before TMT rolling begins.

3.5 TMT Bar Size, Area, Volume & Bundle Reference Chart

The following chart is derived from the plant's reference spreadsheet (Size_and_bundle_chart.xlsx). Calculations use $\pi = 3.14$, Density of iron = 0.00785 g/mm³, and standard bar length = 11 m. The weight per metre is computed using the industry formula: Weight (kg/m) = $D^2 \div 162$, where D is the diameter in mm. Area and Volume are derived from first principles (Area = $\pi \times r^2$; Volume per unit length = Area $\times$ Density).

Area (mm ²) = $\pi \times r^2 = 3.14 \times (D/2)^2$	Volume (g/mm length) = Area $\times$ Density (0.00785)	Weight per metre (kg/m) = $D^2 \div 162$ (simplified formula)
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Size	Dia (mm)	Radius (mm)	Area (mm ²)	Volume (g/mm)	Wt/m (kg)	Wt/11m (kg)	Qty / Bundle	Bundle Wt (kg)	IS 1786 Tol.	+ Max (kg)	Min (kg)
8mm	8	4	50.24	0.3944	0.395	4.345	10	43.45	$\pm 7\%$	46.49	40.41
10mm	10	5	78.5	0.6162	0.617	6.787	7	47.51	$\pm 7\%$	50.84	44.18
12mm	12	6	113.04	0.8874	0.889	9.779	5	48.9	$\pm 5\%$	51.34	46.45
16mm	16	8	200.96	1.5775	1.58	17.38	3	52.14	$\pm 5\%$	54.75	49.53

Size	Dia (mm)	Radius (mm)	Area (mm ²)	Volume (g/mm)	Wt/m (kg)	Wt/11m (kg)	Qty / Bundle	Bundle Wt (kg)	IS 1786 Tol.	+ Max (kg)	Min (kg)
20mm	20	10	314	2.4649	2.469	27.159	2	54.32	±3%	55.95	52.69
25mm	25	12.5	490.62	3.8514	3.858	42.438	1	42.44	±3%	43.71	41.17
32mm	32	16	803.84	6.3101	6.321	69.531	1	69.53	±3%	71.62	67.45
44mm	44	22	1519.76	11.93	11.951	131.461	1	131.46	N/A	-	-

This assignment report is a true summary of observations and learnings from the internship at Hindupur steels and alloys (P) Ltd. , SIPWeek 1&2

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